

Introduction to Real Analysis – Homework 1

Elementary and Jordan measure

Notation. Write $m(A)$ for the measure of an elementary set A . For a bounded set $E \subset \mathbb{R}^d$, recall

$$m_{*,J}(E) = \sup_{\substack{A \subset E \\ A \text{ elementary}}} m(A), \quad m^{*,J}(E) = \inf_{\substack{E \subset B \\ B \text{ elementary}}} m(B).$$

The set E is Jordan measurable when these two quantities agree.

1. **Inclusion–exclusion.** For elementary sets $E, F \subset \mathbb{R}^d$, prove

$$m(E \cup F) + m(E \cap F) = m(E) + m(F).$$

2. **Inner and outer approximation.** Let $E \subset \mathbb{R}^d$ be bounded. Prove that E is Jordan measurable if and only if, for every $\varepsilon > 0$, there exist elementary sets A, B such that

$$A \subset E \subset B, \quad m(B \setminus A) < \varepsilon.$$

Prove both directions directly from the definitions of $m_{*,J}$ and $m^{*,J}$.

3. **Jordan null sets.** A bounded set is *Jordan null* if its Jordan outer measure is zero. Prove that:

- (a) Every subset of a Jordan null set is Jordan null.
- (b) A finite union of Jordan null sets is Jordan null.
- (c) The corresponding assertion for countable unions is false, even when the union is bounded.

4. **Finite operations on Jordan measurable sets.** Suppose $E, F \subset \mathbb{R}^d$ are Jordan measurable. Use Problem 2 to prove that $E \cup F$ and $E \setminus F$ are Jordan measurable.

Hint for the difference: start with $A_E \subset E \subset B_E$ and $A_F \subset F \subset B_F$. Which approximations should you subtract?

5. **A convergent sequence.** Let $x_n \rightarrow x$ in $\mathbb{R}^d$. Prove that

$$\{x\} \cup \{x_n : n \in \mathbb{N}\}$$

is Jordan null.

6. **Challenge: the graph of a continuous function.** Let $a < b$ and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Prove that its graph

$$G_f = \{(x, f(x)) : a \leq x \leq b\}$$

is Jordan null in $\mathbb{R}^2$. *Hint:* use uniform continuity to cover the graph by finitely many thin rectangles.